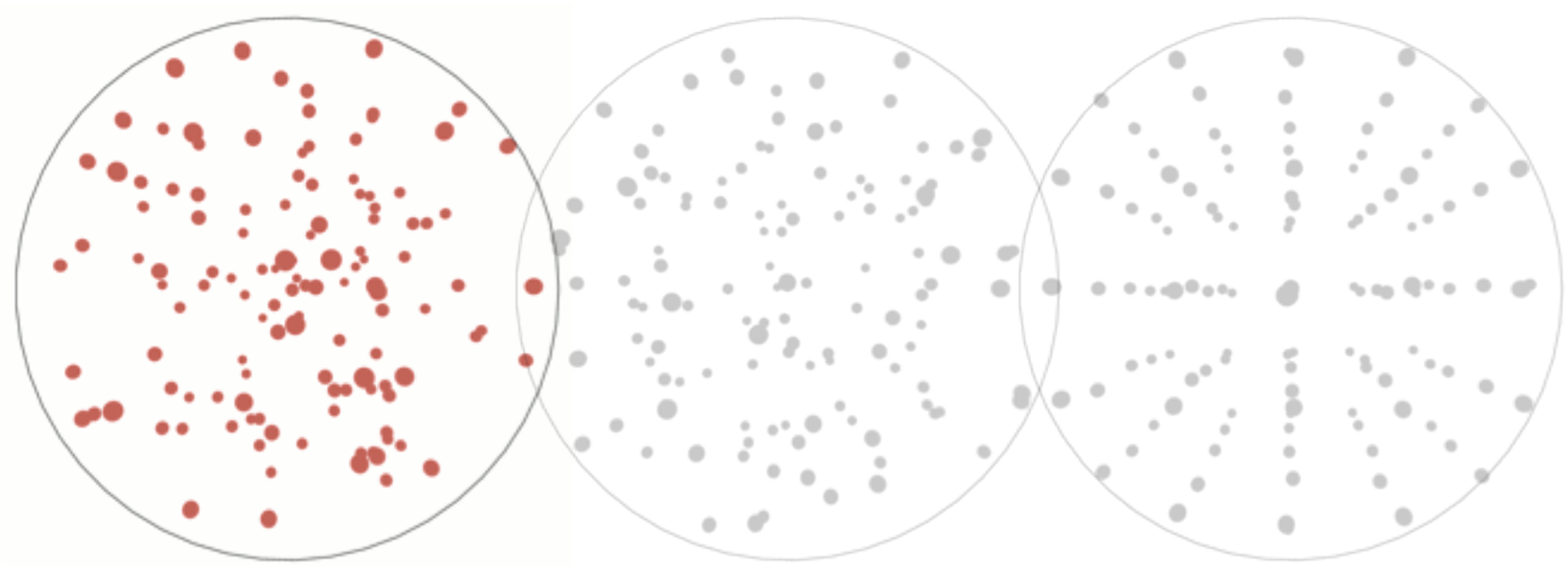
Torsten Scholak, Thomas Wellens, and Andreas Buchleitner:

Exciton transport in ordered and disordered samples of cold Rydberg atoms

QOS
Quantum Optics and Statistics
Albert-Ludwigs-Universität Freiburg

motivation



outline

- working conditions
- Hamiltonian
- disorder and localization

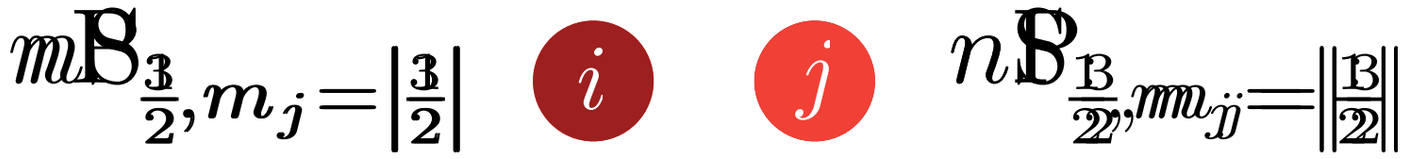
working conditions

frozen gases

- magneto-optical trap, $100\mu\mathrm{K}$
- densities $> 10^7\mathrm{cm}^{-3}$
- excite Rb atoms to Rydberg states
- amorphous solid
- radiation-less long-range interactions

Hamiltonian

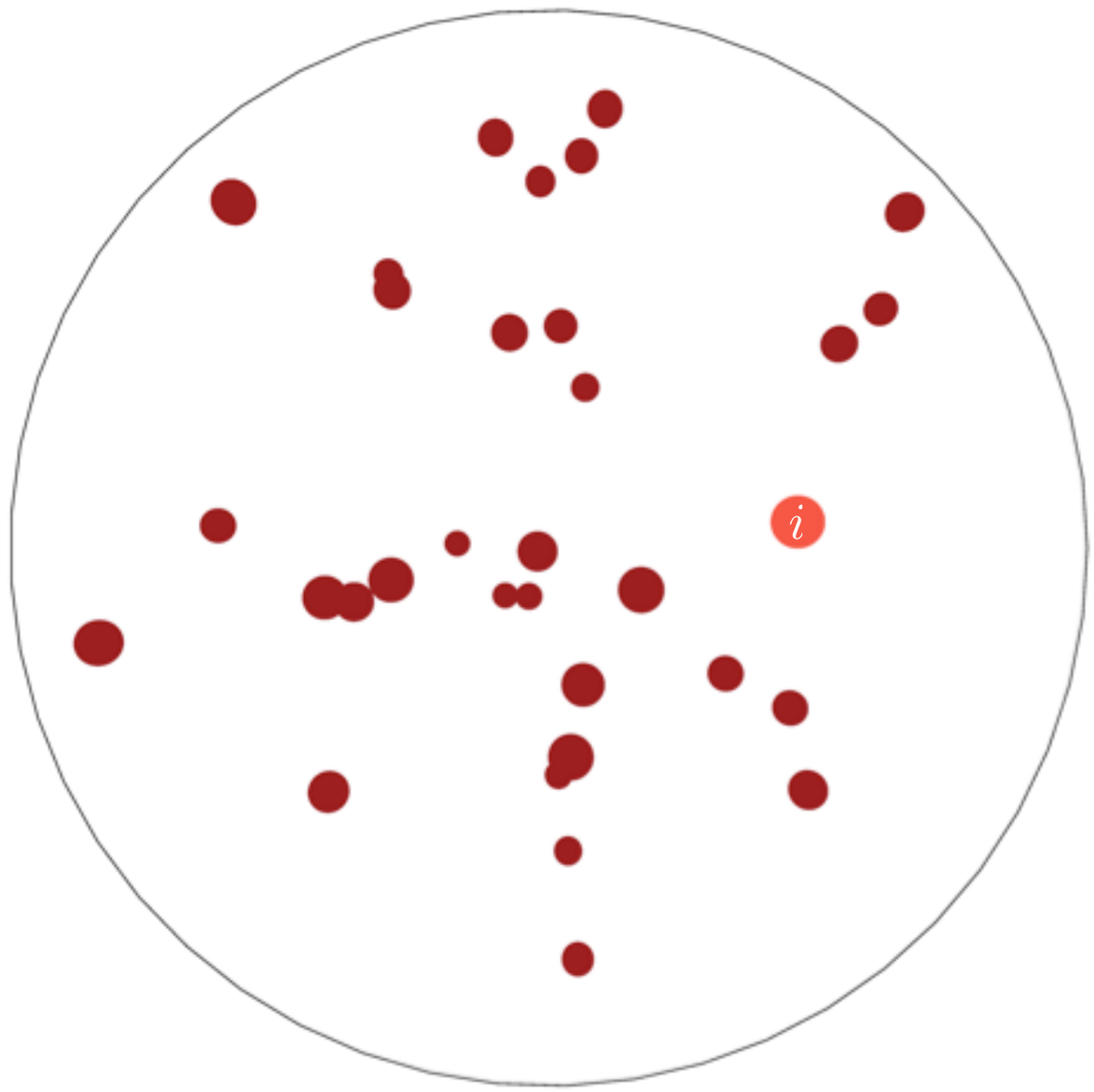
interactions



$$V_{ij} \propto \mathcal{A}(\vartheta_{ij}, \varphi_{ij})\, R_{ij}^{-3}$$

Hamiltonian

one P excitation



$$|\mathrm{S\,S\,S}\ldots\mathrm{P}\ldots\mathrm{S}\rangle := |i\rangle$$

Hamiltonian

tight-binding model

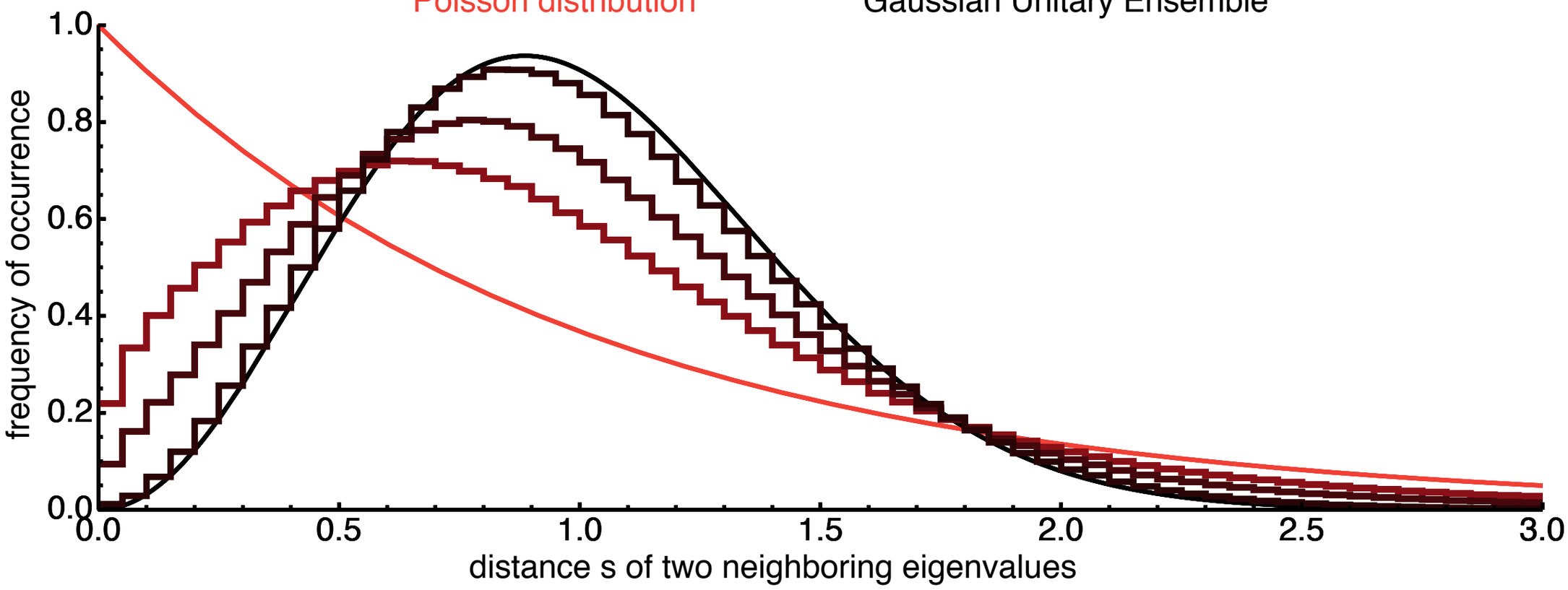
$$H_{ij} = \langle i|\hat{H}|j\rangle = (\varepsilon_{\mathrm{P}} - \varepsilon_{\mathrm{S}})\delta_{ij} + (1-\delta_{ij})V_{ij}$$

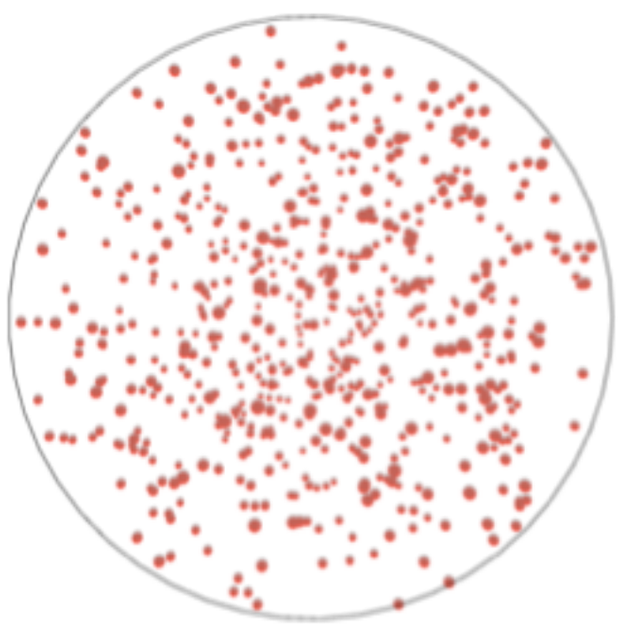
disorder and localization

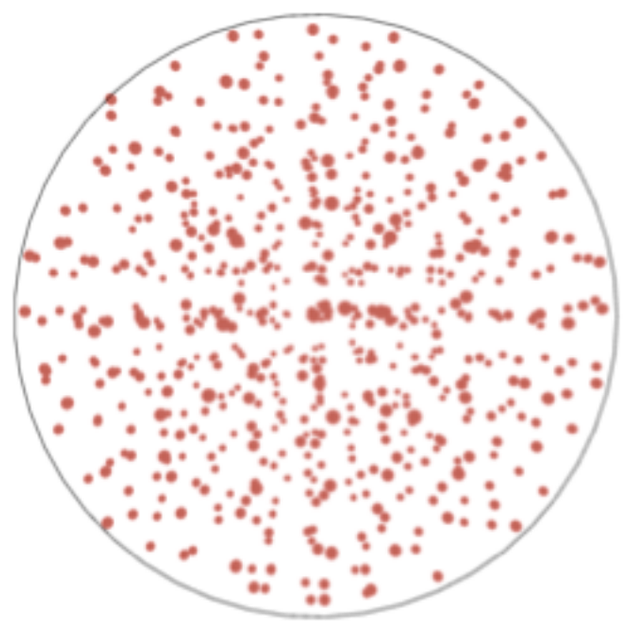
level spacings

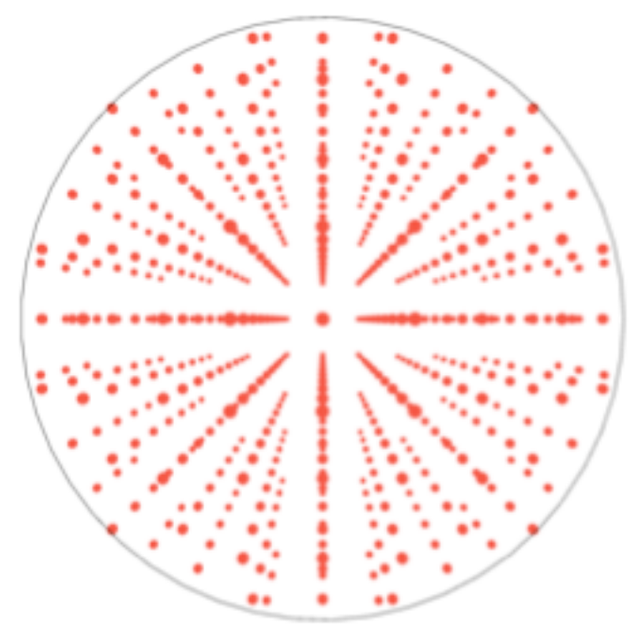
localization ⊥ level repulsion

Poisson distribution — Gaussian Unitary Ensemble





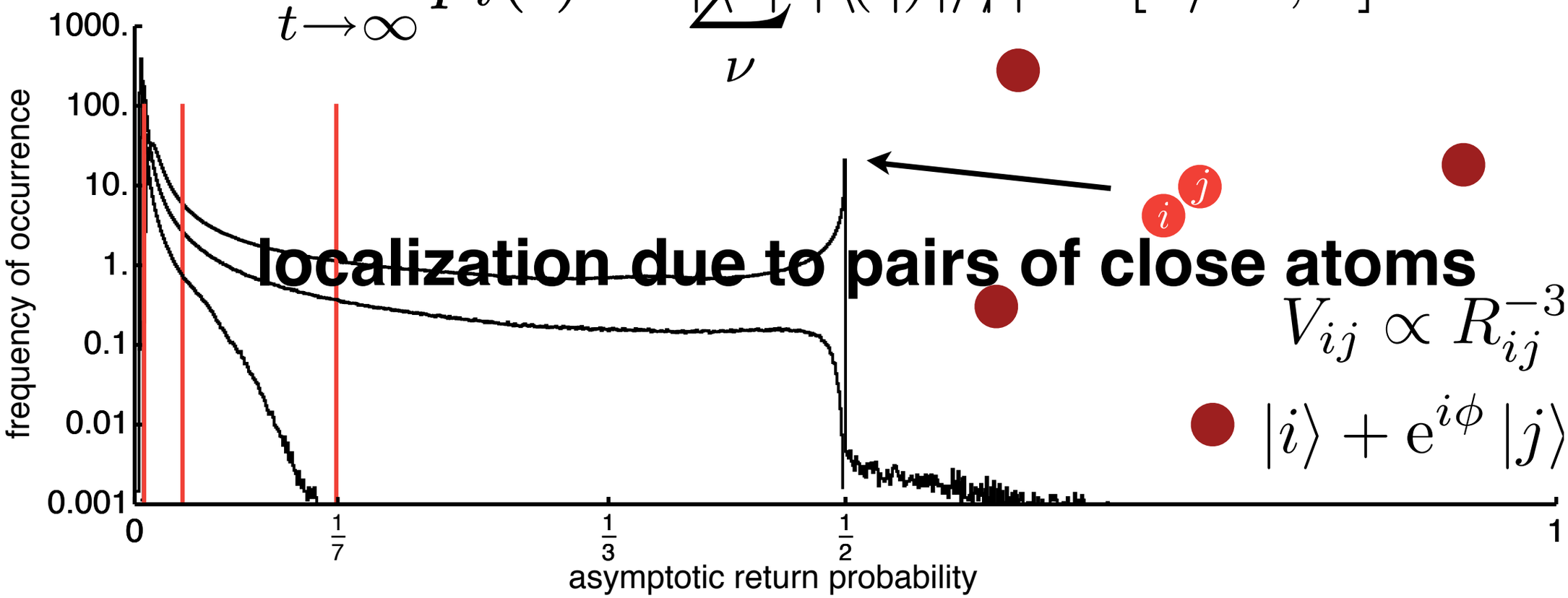




disorder and localization

return probability

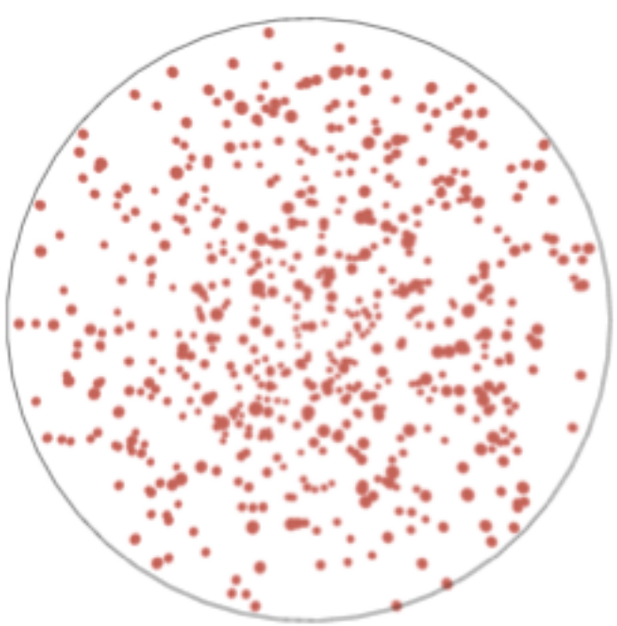
$$\lim_{t\to\infty} p_i(t) = \sum_\nu |\langle i|\psi_\nu\rangle|^2 \in [1/N, 1]$$

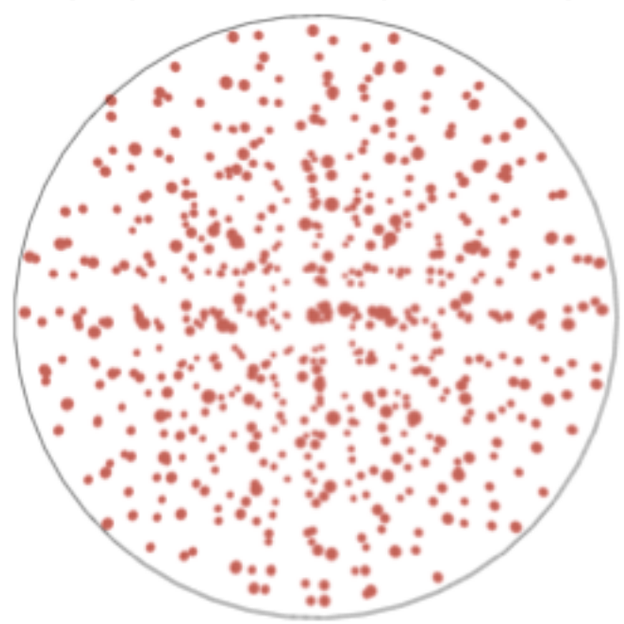


localization due to pairs of close atoms

$$V_{ij} \propto R_{ij}^{-3}$$

$$|i\rangle + e^{i\phi}\,|j\rangle$$





abridged

- sensitivity to spatial disorder
- localization $\leftrightarrow$ proximity of atoms
- experimental observation

Thanks for listening

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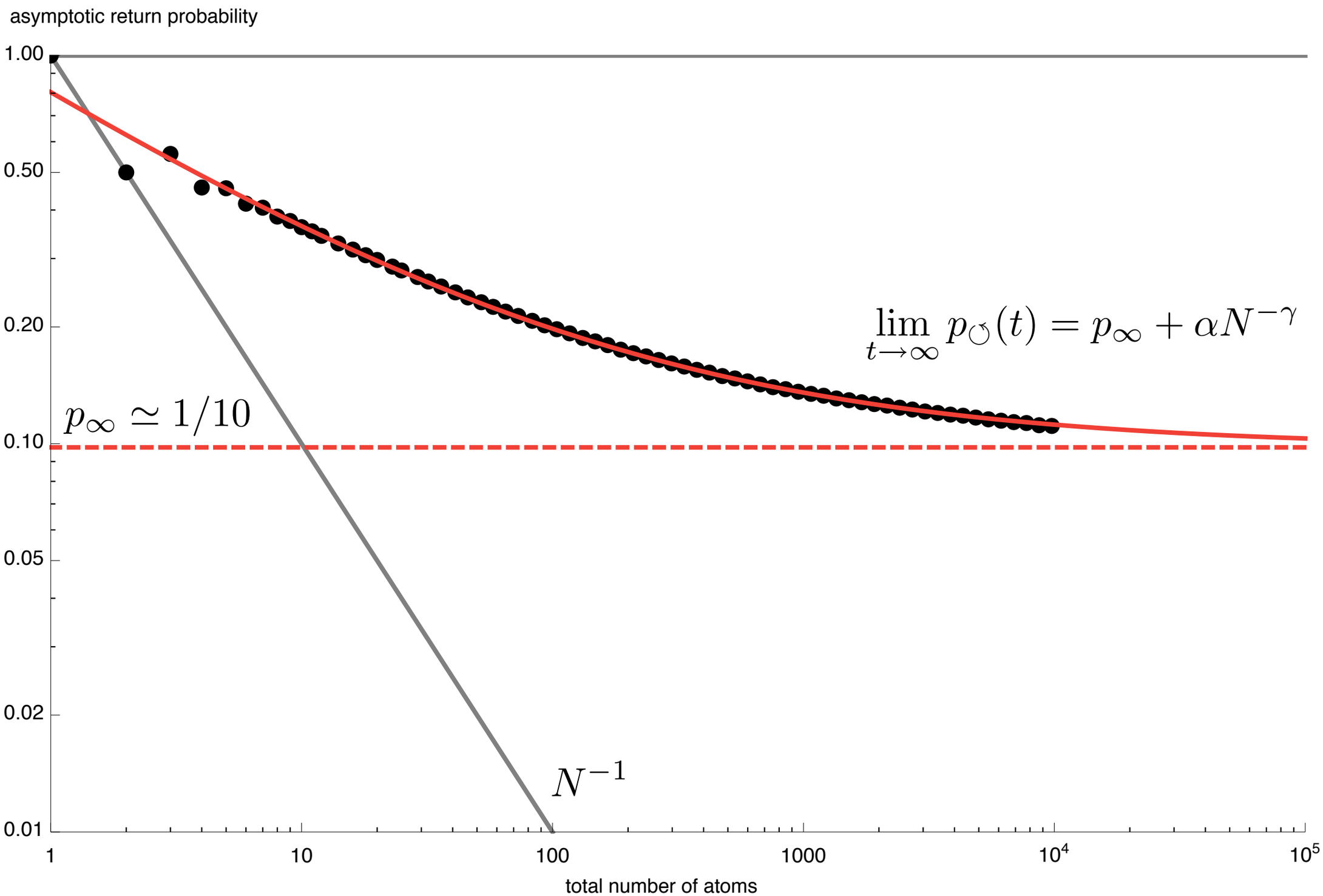
disorder and localization

one last thing...